RRMM — Release Record Management Module

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OOF® Origin Open Foundation®

Independent Methodological Authority

Architecture: SLGA™ — Software Lifecycle Governance Architecture

Parent Standard: SRLS — Software Release Standard

Category: Governance & Enforcement

Subcategory: Software Lifecycle Governance

Type: Core Module

Governed Space: Software Release Record Governance

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Canonical Definition

RRMM — Release Record Management Module is the Core Module within SRLS that governs how software release records are established, structured, associated, preserved, updated, verified, superseded, and maintained as traceable evidence of identifiable software releases and their applicable governance states throughout the software lifecycle.

RRMM establishes the methodological conditions under which a software

release record may represent the relationship between an identifiable release candidate, its artifact and configuration references, readiness assessment, authorization decision, applicable restrictions, release events, and subsequent lifecycle changes.

It governs the integrity, completeness, provenance, consistency,

accessibility, versioning, and continuity of release governance records.

RRMM distinguishes the existence of a release record from the completeness

of its evidence, the recording of an authorization decision from the legitimacy of that decision, and the preservation of historical release information from the continued applicability of its original governance conclusions.

The module ensures that release records remain usable for subsequent deployment, change management, incident investigation, withdrawal, retirement, and lifecycle control without independently replacing the governance mechanisms responsible for those decisions.

Its principal governed output is the Software Release Record Profile.

Release Record Exists ≠ Release Governance Evidence Complete.

Release Recorded ≠ Release Authorized.

Release Record Updated ≠ Underlying Release State Changed.

Historical Release Record Preserved ≠ Historical Authorization Still

Valid.

Artifact Reference Recorded ≠ Artifact Identity Verified.

Operational Role

RRMM governs the establishment and continuity of software release records.

Its operational role is to establish:

• Release record identity

• Release record purpose

• Software product association

• Release candidate association

• Artifact and version references

• Configuration and dependency references

• Readiness assessment references

• Authorization decision references

• Release restrictions

• Release event references

• Release record boundaries

• Record provenance

• Record completeness

• Record consistency

• Evidence sufficiency

• Record integrity

• Record versioning

• Historical state preservation

• Record correction

• Record supersession

• Record accessibility

• Record retention conditions

• Record transfer

• Release withdrawal references

• Deployment handoff

• Lifecycle traceability

RRMM does not independently establish release readiness, grant release authorization, execute deployment, withdraw released software, or certify the correctness of recorded claims.

Operational Question

How can software release records preserve a reliable, evidence-supported, and traceable representation of what software was prepared, assessed, authorized, released, restricted, changed, or withdrawn, while ensuring that record completeness, provenance, integrity, historical continuity, and current governance applicability remain distinguishable throughout human-operated, automated, and autonomous software lifecycles?

Position in Architecture Governance Flow

SLGA™ Governance Flow

Software Requirements → Software Creation → Software Dependencies →

Software Configuration → Software Quality → Software Security → Software Release → Software Deployment → Software Change Management → Software Retirement

Cross-cutting:

Software Lifecycle Control

SRLS Core Module Flow

RCMM — Release Candidate Management Module

2. RRM — Release Readiness Module 3. RAM — Release Authorization Module 4. RRMM — Release Record Management Module 5. RWM — Release Withdrawal Module

RRMM occupies the fourth Core Module position within SRLS.

It receives release candidate identity, readiness conclusions, and

authorization decisions from the preceding SRLS modules.

Its role is to establish a coherent release governance record that preserves the relationship between these states without merging them into a single unsupported claim of release legitimacy.

RRMM supports downstream deployment and lifecycle governance by providing

identifiable, traceable release references.

Release Candidate Profile → Release Readiness Profile → Release

Authorization Profile → Release Record Identity → Record Scope → Artifact & Configuration References → Evidence Association → Record Integrity Assessment → Record Classification → Software Release Record Profile → Deployment / Withdrawal / Lifecycle Governance Handoff

• Governed Space

Software Release Record Governance

This governed space concerns the evidence-supported representation and

preservation of software release governance information.

It includes:

• Release record identification

• Record scope and boundaries

• Release candidate association

• Artifact identification references

• Readiness and authorization references

• Release event recording

• Record provenance

• Evidence association

• Record completeness

• Record integrity

• Record consistency

• Record versioning

• Record correction

• Historical state preservation

• Record supersession

• Record access and retention

• Governance handoffs

RRMM does not independently determine whether the software is technically correct, secure, ready, authorized, deployed, or operationally acceptable.

Governed Object

The primary Governed Object is:

A Software Release Record State representing the evidence-supported

relationship between an identifiable software release, its candidate and artifact references, readiness and authorization states, applicable release conditions, recorded release events, associated evidence, historical changes, and current record integrity.

The governed state must preserve:

Software Product ID

Release Record ID

Release Candidate ID

Artifact Reference

Configuration Reference

Readiness Profile Reference

Authorization Profile Reference

Release Event Reference

Release Scope

Release Restrictions

Evidence References

Record Provenance

Record Completeness State

Record Integrity State

Record Classification

Historical State References

Version

Scope

RRMM governs:

• Release record creation

• Release identity association

• Release candidate references

• Artifact and configuration references

• Readiness and authorization associations

• Release event recording

• Record provenance

• Record completeness

• Evidence sufficiency

• Record consistency

• Record integrity

• Record updates

• Corrections and supersession

• Historical record preservation

• Record accessibility

• Record retention

• Record transfer

• Deployment and withdrawal references

• Lifecycle governance handoffs

RRMM does not independently perform software verification, authorize release decisions, or establish the operational state of deployed software.

Foundational Governance Principles8.1 Record Existence Principle

Release Record Exists ≠ Release Governance Evidence Complete. 8.2 Authorization Separation Principle

Release Recorded ≠ Release Authorized. 8.3 Record–Reality Separation Principle

Release Record Updated ≠ Underlying Release State Changed. 8.4 Historical Validity Principle

Historical Release Record Preserved ≠ Historical Authorization Still

Valid. 8.5 Artifact Identity Principle

Artifact Reference Recorded ≠ Artifact Identity Verified. 8.6 Provenance Principle

Release Information Recorded ≠ Release Information Proven. 8.7 Completeness Principle

Required Fields Populated ≠ Release Record Substantively Complete. 8.8 Integrity Principle

Release Record Accessible ≠ Release Record Integrity Established. 8.9 Version Continuity Principle

Release Record Revised ≠ Previous Release Record Erased. 8.10 Lifecycle Traceability Principle

Release History Available ≠ Complete Software Lifecycle Traceability

Established.

Release Record Identity

Every governed release record must establish:

Release Record ID

Software Product ID

Release Candidate ID

Release Identifier

Record Purpose

Record Scope

Record Origin

Record Creation Time

Record Version

Record Classification

Record Identity Integrity

Release Name Recorded ≠ Unique Release Identity Established.

Release records must remain distinguishable across products, candidate revisions, artifacts, and release channels.

Record Purpose

Release record purposes may include:

Release Governance Documentation

Release Evidence Preservation

Release Authorization Traceability

Release Distribution Traceability

Deployment Reference

Release Withdrawal Reference

Lifecycle Investigation

Historical Governance Reconstruction

A record created for one purpose must not automatically be treated as sufficient for every other purpose.

Purpose Principle

Release Record Suitable for Distribution Tracking ≠ Release Record

Sufficient for Authorization Audit.

Release Record Boundary

The record boundary defines the software release information represented by

the record.

Relevant dimensions include:

Software Product Boundary

Release Candidate Boundary

Artifact Boundary

Configuration Boundary

Dependency Boundary

Release Channel Boundary

Authorization Boundary

Evidence Boundary

Temporal Boundary

Lifecycle Boundary

Boundary Integrity

Release Record Covers One Candidate ≠ Release Record Covers Every Related

Artifact.

Software Product Association

RRMM must preserve the relationship between the release record and its software product.

Relevant elements include:

Software Product ID

Product Name

Product Version Reference

Product Scope

Product Ownership Reference

Product Lifecycle Reference

Associated Release Candidates

Product Association Principle

Software Product Identified ≠ Exact Released Artifact Identified.

Release Candidate Association

RRMM receives the governed candidate identity established by RCMM.

Relevant elements include:

Release Candidate ID

Candidate Revision

Candidate Baseline

Candidate Composition Reference

Candidate Status

Candidate Evidence Reference

Candidate Supersession Reference

Candidate Association Principle

Release Candidate Referenced ≠ Candidate Identity and Composition

Verified.

Artifact Identity References

RRMM preserves references to the software artifacts associated with a

release.

Relevant elements include:

Artifact ID

Artifact Type

Artifact Version

Artifact Hash or Equivalent Identifier

Build Reference

Artifact Origin

Artifact Location Reference

Artifact Integrity Evidence

Artifact identity verification remains dependent on the applicable

configuration and integrity mechanisms.

Artifact Reference Principle

Version Label Equality ≠ Artifact Equivalence.

Configuration References

RRMM preserves the configuration context relevant to the release.

Relevant elements include:

Configuration Baseline ID

Build Configuration Reference

Runtime Configuration Reference Where Applicable

Environment Assumptions

Configuration Restrictions

Configuration Evidence

Configuration Version

Configuration Principle

Release Artifact Identified ≠ Applicable Configuration Context Fully

Established.

Dependency References

RRMM preserves material dependency references relevant to release

reconstruction.

Relevant elements include:

Dependency Baseline ID

Component Inventory Reference

Dependency Origin Reference

Dependency Version Reference

Dependency Restriction

Dependency Risk Reference

Dependency Evidence

Dependency Reference Principle

Dependency Manifest Attached ≠ Complete Release Dependency Reality

Established.

Readiness Assessment References

RRMM must preserve the applicable readiness determination.

Relevant elements include:

Readiness Assessment ID

Readiness Classification

Readiness Assessment Time

Assessment Boundary

Assessment Evidence

Unresolved Findings

Applicable Restrictions

Reassessment Requirements

RRMM records the readiness state established by RRM.

It does not independently establish readiness.

Readiness Record Principle

Readiness Conclusion Recorded ≠ Readiness Conclusion Independently

Verified by RRMM.

Authorization Decision References

RRMM must preserve the applicable release authorization decision.

Relevant elements include:

Release Authorization ID

Authorization Classification

Authorization Authority Reference

Decision Actor Reference

Decision Time

Authorized Release Scope

Authorization Restrictions

Validity Conditions

Authorization Evidence Reference

RRMM records the authorization state established by RAM.

It does not independently grant authorization.

Authorization Record Principle

Authorization Decision Referenced ≠ Authorization Legitimacy Independently

Established.

Release Event Recording

RRMM may preserve release events associated with the software release.

Relevant elements include:

Release Event ID

Release Event Type

Release Candidate ID

Artifact Reference

Release Channel

Event Actor

Event Time

Event Evidence

Event Classification

Events must remain distinguishable from authorization decisions.

Release Event Principle

Release Event Recorded ≠ Release Event Authorized.

Release Distribution References

Where applicable, RRMM preserves distribution-related references.

Relevant elements include:

Distribution Event ID

Distribution Channel

Distributed Artifact Reference

Distribution Time

Distribution Actor

Recipient Scope

Distribution Evidence

Distribution Restrictions

RRMM does not independently establish successful delivery to every intended recipient.

Distribution Principle

Software Published to a Repository ≠ Software Delivered to Every Intended

Recipient.

Release Restrictions

RRMM preserves restrictions applicable to the recorded release.

Relevant restrictions include:

Restricted Distribution

Limited Recipient Scope

Configuration Restrictions

Environment Restrictions

Validity Period Restrictions

Conditional Release Requirements

Exception-Related Restrictions

Withdrawal Conditions

Restriction Integrity

Release Restrictions Recorded ≠ Release Restrictions Operationally

Enforced.

Release Record Provenance

RRMM governs the traceability of release record origins and updates.

Relevant elements include:

Record Origin

Record Creator

Creation Method

Source Systems

Referenced Evidence

Record Creation Time

Modification History

Responsible Record Function

Record Version

Provenance Integrity

Release Record Source Identified ≠ Release Record Content Authenticated.

Record Evidence Association

RRMM must associate release record claims with relevant evidence.

Evidence may include:

Candidate Identity Evidence

Artifact Integrity Evidence

Configuration Evidence

Dependency Evidence

Readiness Evidence

Authorization Evidence

Release Event Evidence

Distribution Evidence

Record Modification Evidence

Withdrawal References

Evidence Association Principle

Evidence Linked ≠ Evidence Relevant or Sufficient.

Record Completeness

Record completeness concerns whether the record contains the information

required for its defined governance purpose.

Relevant dimensions include:

Identity Completeness

Candidate Reference Completeness

Artifact Reference Completeness

Readiness Reference Completeness

Authorization Reference Completeness

Release Event Coverage

Evidence Coverage

Restriction Coverage

Historical State Coverage

Completeness Integrity

All Mandatory Fields Present ≠ All Recorded Claims Substantively

Supported.

Record Consistency

RRMM evaluates whether the record contains contradictory or incompatible

references.

Relevant dimensions include:

Candidate Identity Consistency

Artifact Version Consistency

Configuration Consistency

Readiness Reference Consistency

Authorization Scope Consistency

Release Event Consistency

Timestamp Consistency

Historical State Consistency

Consistency Principle

Release Record Internally Consistent ≠ Release Record Factually Correct.

Record Integrity

RRMM governs whether the release record remains protected against

untraceable or unauthorized alteration.

Relevant elements include:

Record Integrity Reference

Modification Authorization

Change Traceability

Version History

Integrity Verification Evidence

Access Restrictions

Correction History

Supersession History

Integrity Principle

Record Unchanged ≠ Record Originally Accurate.

Record Authenticity

Record authenticity concerns whether the record's claimed origin and

modifications are sufficiently supported.

Relevant elements include:

Origin Identity

Record Creator Identity

Source System Reference

Creation Evidence

Modification Evidence

Authentication Mechanism Reference

Known Authenticity Limitations

Authenticity Principle

Record Authenticated ≠ Every Claim Within the Record Validated.

Record Evidence Sufficiency

RRMM evaluates whether the available evidence adequately supports the

record's intended governance use.

Relevant dimensions include:

Evidence Coverage

Evidence Provenance

Evidence Relevance

Evidence Consistency

Evidence Freshness

Evidence Accessibility

Evidence Integrity

Known Evidence Gaps

Sufficiency Principle

Release Evidence Available ≠ Release Evidence Sufficient for Every

Lifecycle Decision.

Missing Record Evidence

RRMM must preserve missing or unavailable evidence.

Examples include:

Missing Artifact Identity Evidence

Missing Configuration Reference

Missing Readiness Assessment

Missing Authorization Decision

Missing Release Event Evidence

Missing Distribution Reference

Missing Record Modification History

Unavailable Historical Record

Missing Evidence Principle

Release Information Missing ≠ Release Condition Satisfied.

Record Uncertainty

Record uncertainty may arise from:

Unverified Artifact Identity

Incomplete Release History

Conflicting Authorization References

Unknown Release Actor

Uncertain Distribution Scope

Missing Historical Evidence

Inconsistent Timestamps

Unverified External Records

Uncertainty Principle

Record Uncertainty Acknowledged ≠ Record Uncertainty Resolved.

Record Updates

RRMM governs modifications to existing release records.

Relevant elements include:

Record Update ID

Affected Release Record

Update Purpose

Update Actor

Update Authority Reference

Previous Value

Updated Value

Update Evidence

Update Time

Updated Version

Update Integrity

Release Record Corrected ≠ Original Release Event Changed.

Record Correction

A record correction addresses inaccurate, incomplete, or misleading

recorded information.

Relevant elements include:

Correction ID

Affected Field

Original Recorded Value

Corrected Value

Correction Reason

Supporting Evidence

Correction Authority Reference

Correction Time

Historical Reference

Corrections must preserve the distinction between historical recording and later correction.

Correction Principle

Historical Record Inaccurate ≠ Historical Record May Be Silently Rewritten.

Record Supersession

RRMM governs the relationship between earlier and later release records.

Relevant elements include:

Original Record ID

Superseding Record ID

Supersession Reason

Supersession Scope

Effective Time

Evidence Reference

Historical Relationship

Supersession must not erase the historical record.

Supersession Principle

Release Record Superseded ≠ Previous Release Record Invalid for Historical

Reconstruction.

Record Versioning

Every material record update must preserve version continuity.

Relevant elements include:

Record Version

Previous Version

Version Change Reason

Modified Fields

Modification Actor

Modification Time

Evidence Reference

Supersession Reference

Versioning Principle

Release Record Version Increased ≠ Software Release Version Increased.

Historical Release State Preservation

RRMM preserves historical governance states relevant to the release.

Relevant elements include:

Previous Candidate State

Previous Readiness State

Previous Authorization State

Previous Release Event State

Previous Distribution State

Previous Record Classification

Historical Evidence

State Transition Time

Superseding State

Historical Integrity

Current Release Record State ≠ Complete Historical Release State.

Record Retention

RRMM governs the identification and preservation of applicable release

record retention conditions.

Relevant elements include:

Retention Purpose

Retention Authority Reference

Retention Period

Record Category

Evidence Retention Requirement

Access Conditions

Archival Reference

Disposal Restriction

Retention requirements may arise from applicable organizational, contractual, regulatory, or legal obligations.

RRMM does not independently establish those external obligations.

Retention Principle

Record No Longer Operationally Needed ≠ Record Eligible for Deletion.

Record Accessibility

RRMM governs whether authorized governance functions can access the release record when required.

Relevant elements include:

Record Location

Access Authority

Access Scope

Availability State

Retrieval Method

Record Format

Evidence Accessibility

Historical Version Accessibility

Accessibility Principle

Release Record Stored ≠ Release Record Retrievable When Required.

Record Transfer and Continuity

RRMM preserves release record continuity across organizational, technical,

or custodial changes.

Relevant elements include:

Original Record Custodian

Receiving Record Custodian

Transfer Scope

Transferred Record Identity

Transferred Evidence

Transfer Integrity

Access Continuity

Historical Version Preservation

Transfer Time

Transfer Principle

Release Record Transferred ≠ Record Integrity and Accessibility

Automatically Preserved.

Release Record Classification

RRMM may classify release record states as:

Record Not Established

Record Creation Pending

Record Established

Record Incomplete

Record Evidence-Limited

Record Verification Pending

Record Consistent

Record Conflicted

Record Integrity Unresolved

Record Integrity Supported

Record Correction Required

Record Update Pending

Record Superseded

Record Archived

Record Inaccessible

Record Retention Exception

Unknown

Record Closed

These are release record governance classifications, not independent readiness or authorization classifications.

Classification Integrity

Release Record Classified as Complete ≠ Release Independently Validated as

Ready or Authorized.

Software Release Record Profile

The principal governed output is the:

Software Release Record Profile

It establishes the evidence-supported record state of an identifiable

software release.

The profile preserves:

• Software product identity

• Release record identity

• Release candidate identity

• Artifact and version references

• Configuration references

• Dependency references

• Readiness assessment references

• Authorization decision references

• Release event references

• Distribution references

• Release restrictions

• Record provenance

• Evidence associations

• Record completeness

• Record consistency

• Record integrity

• Record classification

• Record corrections

• Historical versions

• Supersession references

• Retention and accessibility conditions

• Lifecycle governance handoffs

Minimum Profile Fields

At minimum:

Software Release Record Profile ID

Release Record ID

Software Product ID

Release Candidate ID

Release Identifier

Release Record Purpose

Release Record Boundary

Artifact ID

Artifact Version

Artifact Integrity Reference

Build Reference

Configuration Baseline ID

Dependency Baseline ID

Readiness Assessment ID

Readiness Classification

Release Authorization ID

Authorization Classification

Authorization Scope

Authorization Restrictions

Release Event ID

Release Event Type

Release Event Time

Release Event Actor

Distribution Reference

Record Origin

Record Creator

Record Creation Time

Record Evidence Register

Evidence Provenance

Evidence Sufficiency

Missing Evidence

Evidence Conflicts

Record Completeness

Record Consistency

Record Integrity

Record Classification

Correction References

Supersession References

Historical State References

Retention Conditions

Accessibility State

Lifecycle Handoff References

Version

Minimum Implementation Framework

A minimum operational RRMM implementation should contain eight governed layers. Layer 1 — Record Identity and Scope

Establish the release record identity, purpose, boundaries, and software

product association. Layer 2 — Candidate and Artifact Association

Associate the record with the identifiable release candidate, artifact,

build, configuration, and dependency references. Layer 3 — Readiness and Authorization Association

Preserve the applicable readiness assessment and authorization decision

without independently replacing either mechanism. Layer 4 — Release Event and Restriction Recording

Record applicable release events, distribution references, and release

restrictions. Layer 5 — Evidence and Provenance

Establish record provenance, evidence associations, missing evidence, and

evidence sufficiency. Layer 6 — Record Integrity and Classification

Evaluate record completeness, consistency, integrity, and applicable record

classifications. Layer 7 — Historical Continuity

Preserve corrections, versions, supersession relationships, retention

conditions, and historical states. Layer 8 — Governed Output and Lifecycle Handoff

Produce the Software Release Record Profile and provide relevant

references to deployment, withdrawal, change management, retirement, and lifecycle control governance.

Minimum Decision States

At minimum:

Record Not Established

Record Creation Requested

Release Record Identity Established

Record Purpose Defined

Record Boundary Defined

Software Product Identified

Release Candidate Identified

Artifact Reference Established

Configuration Reference Established

Dependency Reference Established

Readiness Reference Established

Authorization Reference Established

Release Event Recorded

Release Restrictions Recorded

Record Evidence Associated

Record Provenance Established

Evidence Sufficiency Review Pending

Record Evidence Sufficient

Record Evidence Insufficient

Record Completeness Review Pending

Record Complete Within Defined Scope

Record Incomplete

Record Consistency Review Pending

Record Consistent

Record Conflicted

Record Integrity Review Pending

Record Integrity Supported

Record Integrity Unresolved

Record Correction Required

Record Correction Completed

Record Update Pending

Record Updated

Record Superseded

Record Archived

Record Inaccessible

Lifecycle Handoff Completed

Unknown

Record Closed

Minimum Evidence Path

Where release record integrity or completeness cannot be sufficiently established:

Software Product → Release Candidate → Artifact & Configuration References

→ Readiness Profile → Authorization Profile → Release Event References → Record Provenance → Evidence Association → Completeness Review → Consistency Review → Integrity Review → Evidence-Limited / Conflicted / Unknown State → Missing Evidence Identification → Evidence Acquisition / Record Correction → Record Reassessment → Updated Record Classification → Software Release Record Profile → Lifecycle Governance Handoff

Missing record evidence must not be replaced by assumed release legitimacy.

Minimum Reassessment Framework

Current Software Release Record Profile → New Release Information or

Evidence → Record Identity Review → Candidate Association Review → Artifact Reference Review → Configuration & Dependency Review → Readiness Reference Review → Authorization Reference Review → Release Event Review → Evidence Provenance Review → Completeness Review → Consistency Review → Integrity Review → Correction / Supersession Assessment → Historical State Preservation → Record Classification Update → Profile Update → Version Update

Reassessment must distinguish:

Release Event Changed

Release Record Changed

Evidence Changed

Readiness Reference Changed

Authorization Reference Changed

Artifact Reference Changed

Record Correction Required

Record Supersession Required

Historical State Preserved

Governance Requirements

A conforming RRMM implementation must satisfy the following requirements. Record Identity and Boundaries

• Establish a unique release record identity

• Identify the software product

• Identify the release candidate

• Define the record purpose

• Define the record boundary

• Preserve the release identifier

• Preserve record creation time

• Preserve record origin

• Preserve record version

• Prevent release names from replacing unique record identities

Candidate, Artifact, and Configuration References

• Preserve the release candidate reference

• Preserve candidate revision information

• Preserve artifact identity references

• Preserve artifact version references

• Preserve artifact integrity references where applicable

• Preserve build references

• Preserve configuration baseline references

• Preserve dependency baseline references

• Preserve known artifact identity limitations

• Prevent version labels from automatically establishing artifact equivalence.

Readiness and Authorization References

• Preserve the applicable readiness assessment reference

• Preserve readiness classification

• Preserve readiness restrictions

• Preserve the applicable authorization reference

• Preserve authorization classification

• Preserve authorization scope

• Preserve authorization restrictions

• Preserve authorization validity references

• Prevent readiness records from automatically establishing release

• Prevent authorization records from independently establishing authorization. authorization legitimacy.

Release Events and Distribution

• Identify relevant release events

• Preserve release event identities

• Preserve event actors

• Preserve event times

• Preserve event evidence

• Preserve release channel references

• Preserve distribution scope where applicable

• Preserve release restrictions

• Prevent release event recording from establishing event

• Prevent repository publication from establishing successful

authorization. distribution to every recipient.

Evidence Governance

• Preserve record evidence provenance

• Preserve evidence source references

• Preserve evidence creation times

• Preserve evidence associations

• Evaluate evidence relevance

• Evaluate evidence sufficiency

• Preserve missing evidence

• Preserve conflicting evidence

• Preserve uncertainty

• Prevent populated record fields from automatically establishing

evidentiary completeness.

Record Completeness and Integrity

• Evaluate record completeness

• Preserve completeness boundaries

• Evaluate record consistency

• Preserve conflicting references

• Evaluate record integrity

• Preserve modification traceability

• Preserve applicable authentication references

• Identify unauthorized or unexplained alterations

• Prevent internal consistency from establishing factual correctness

• Prevent record accessibility from automatically establishing record

integrity.

Record Updates and Historical Continuity

• Preserve material record updates

• Preserve update actors

• Preserve update authority references

• Preserve correction reasons

• Preserve original recorded values

• Preserve corrected values

• Preserve supersession relationships

• Preserve historical record versions

• Preserve state transition times

• Prevent corrections from silently rewriting historical release

• Prevent record supersession from erasing historical records

• Prevent record version changes from being confused with software events. release version changes.

Retention, Accessibility, and Transfer

• Identify applicable record retention conditions

• Preserve retention authority references

• Preserve archival references

• Preserve authorized access conditions

• Preserve record retrieval capability

• Preserve historical version accessibility where required

• Preserve record transfer identity

• Preserve transfer integrity evidence

• Prevent record storage from automatically establishing

• Prevent record transfer from automatically establishing integrity retrievability. continuity.

Architecture Boundaries

• Maintain RCMM candidate governance boundaries

• Maintain RRM readiness governance boundaries

• Maintain RAM authorization governance boundaries

• Maintain RWM withdrawal governance boundaries

• Maintain SCFS configuration governance boundaries

• Maintain SDS dependency governance boundaries

• Maintain SDPS deployment governance boundaries

• Maintain SCMS change governance boundaries

• Maintain SRTS retirement governance boundaries

• Maintain SLCS lifecycle control boundaries

• Prevent RRMM from independently establishing release readiness

• Prevent RRMM from independently granting release authorization

• Prevent RRMM from independently establishing deployment permission

• Prevent RRMM from independently certifying software correctness or

• Produce a Software Release Record Profile

security.

Governance Logic

Canonical RRMM logic:

Software Product Identity → Release Record Identity → Record Purpose →

Record Boundary → Release Candidate Reference → Artifact & Version References → Configuration & Dependency References → Readiness Assessment Reference → Authorization Decision Reference → Release Event References → Distribution & Restriction References → Record Provenance → Evidence Association → Evidence Sufficiency → Record Completeness → Record Consistency → Record Integrity → Record Classification → Software Release Record Profile → Historical State Preservation → Deployment / Withdrawal / Lifecycle Governance Handoff

Governance Outputs

RRMM may produce:

Software Release Record Profile

Release Record Identity

Release Record Purpose

Release Record Boundary

Software Product Association

Release Candidate Association

Artifact Identity Reference

Artifact Version Reference

Build Reference

Configuration Baseline Reference

Dependency Baseline Reference

Readiness Assessment Reference

Authorization Decision Reference

Release Event Register

Distribution Reference

Release Restriction Register

Record Provenance Register

Record Evidence Register

Evidence Sufficiency State

Missing Evidence Record

Evidence Conflict Record

Record Completeness Assessment

Record Consistency Assessment

Record Integrity Assessment

Record Authenticity Reference

Record Classification

Record Update Register

Record Correction Register

Record Supersession Reference

Historical Record Register

Record Retention Reference

Record Accessibility State

Record Transfer Reference

Lifecycle Governance Handoff

Version

Practical ApplicationsAutonomous Software Development

Preserving traceable release records when AI agents create, assess, authorize, and distribute software through multiple automated tools. Enterprise Release Management

Maintaining coherent release governance records across development,

testing, security, operations, and management functions. Continuous Integration and Delivery

Associating automated build artifacts, readiness results, authorization

decisions, and release events without confusing technical pipeline execution with governance approval. Software Supply Chain Governance

Preserving artifact, configuration, dependency, and origin references

needed to reconstruct software release composition. Critical Infrastructure Software

Maintaining reliable release histories for systems where changes may affect essential operational functions. Incident Investigation and Withdrawal

Providing release evidence that supports identification of affected artifacts, release events, authorization conditions, and historical states.

Use Case 1 — An AI Software Factory Produces a Release Without a

Reconstructable Record

An organization operates an autonomous software factory.

Multiple AI agents generate source code, modify dependencies, execute

builds, perform tests, and prepare release candidates.

The software factory successfully produces a functioning application.

A release authorization process approves the candidate.

The software is subsequently published.

However, the organization discovers that its release record contains only a version label and a successful pipeline status.

The record does not preserve the exact artifact identity, applicable configuration baseline, dependency references, readiness assessment, or authorization decision.

RRMM evaluates:

Release Record Identity → Candidate Association → Artifact Identity →

Configuration & Dependency References → Readiness Reference → Authorization Reference → Evidence Sufficiency → Record Classification

The assessment identifies:

• A release event occurred

• A release record exists

• The exact released artifact cannot be sufficiently reconstructed

• Readiness and authorization references are incomplete

• The record cannot fully support subsequent governance investigation

The appropriate classification is:

Record Incomplete — Evidence-Limited.

The critical distinctions are:

Release Record Exists ≠ Release Governance Evidence Complete.

Successful Pipeline ≠ Complete Release Evidence.

Version Label ≠ Verified Artifact Identity.

The result: RRMM exposes the difference between successful automated

software delivery and reconstructable release governance.

• Use Case 2 — A Release Record Is Silently Modified After an Incident

An organization distributes a software release.

Several weeks later, an operational incident reveals that the released software contained a configuration inconsistent with the intended release conditions.

During the investigation, a release administrator modifies the original

release record to reflect the configuration that should have been used.

The modification overwrites the previous value.

No correction reason, modification time, or historical version is preserved.

The current release record appears consistent with the intended

configuration.

However, it no longer accurately represents the information recorded at the

time of release.

RRMM evaluates:

Original Record → Recorded Configuration → Modification Event → Correction

Evidence → Historical State → Record Integrity

The assessment identifies:

• The original record was modified

• The original recorded value was not preserved

• The modification cannot be sufficiently reconstructed

• The current record may conceal the historical discrepancy

• Record integrity is not sufficiently established

The appropriate classification is:

Record Integrity Unresolved — Historical Evidence Incomplete.

The critical distinctions are:

Release Record Corrected ≠ Original Release Event Changed.

Current Record Consistent ≠ Historical Record Accurate.

Record Updated ≠ Record Integrity Preserved.

The result: RRMM requires corrections to remain traceable rather than

allowing historical release evidence to be silently rewritten.

Use Case 3 — A Security Withdrawal Requires Identification of Affected

Releases

A software supplier discovers a security vulnerability affecting a

third-party library.

The vulnerable library was used in several software releases.

Some releases included the library directly.

Others inherited it through transitive dependencies.

The supplier must determine which releases may be affected.

Its release records contain software version names but do not consistently

preserve dependency baselines or artifact composition references.

The organization cannot immediately establish which distributed artifacts

contain the vulnerable component.

RRMM evaluates:

Release Record Identity → Artifact References → Dependency Baseline

References → Historical Release Composition → Evidence Coverage → Affected Release Identification

The assessment identifies:

• Release records are available

• Dependency composition evidence is incomplete

• Version names do not sufficiently establish affected artifact

• The organization cannot reliably reconstruct the complete affected

• Additional evidence acquisition is required

composition. release population.

The appropriate classification is:

Record Evidence-Limited — Dependency Traceability Incomplete.

The critical distinctions are:

Release History Available ≠ Dependency Exposure Reconstructable.

Software Version Known ≠ Component Composition Known.

Record Completeness for Distribution ≠ Record Completeness for Security

Withdrawal.

The result: RRMM demonstrates why release records must preserve sufficient artifact and dependency traceability before an incident creates an urgent need for reconstruction.

Critical Distinctions

Release Record Exists ≠ Release Governance Evidence Complete.

Release Recorded ≠ Release Authorized.

Release Record Updated ≠ Underlying Release State Changed.

Historical Release Record Preserved ≠ Historical Authorization Still

Valid.

Artifact Reference Recorded ≠ Artifact Identity Verified.

Release Information Recorded ≠ Release Information Proven.

Required Fields Populated ≠ Release Record Substantively Complete.

Release Record Accessible ≠ Release Record Integrity Established.

Release Record Revised ≠ Previous Release Record Erased.

Release History Available ≠ Complete Software Lifecycle Traceability

Established.

Release Name Recorded ≠ Unique Release Identity Established.

Release Record Suitable for Distribution Tracking ≠ Release Record

Sufficient for Authorization Audit.

Release Record Covers One Candidate ≠ Release Record Covers Every Related

Artifact.

Software Product Identified ≠ Exact Released Artifact Identified.

Release Candidate Referenced ≠ Candidate Identity and Composition

Verified.

Version Label Equality ≠ Artifact Equivalence.

Release Artifact Identified ≠ Applicable Configuration Context Fully

Established.

Dependency Manifest Attached ≠ Complete Release Dependency Reality

Established.

Readiness Conclusion Recorded ≠ Readiness Conclusion Independently

Verified by RRMM.

Authorization Decision Referenced ≠ Authorization Legitimacy Independently

Established.

Release Event Recorded ≠ Release Event Authorized.

Software Published to a Repository ≠ Software Delivered to Every Intended

Recipient.

Release Restrictions Recorded ≠ Release Restrictions Operationally

Enforced.

Release Record Source Identified ≠ Release Record Content Authenticated.

Evidence Linked ≠ Evidence Relevant or Sufficient.

All Mandatory Fields Present ≠ All Recorded Claims Substantively

Supported.

Release Record Internally Consistent ≠ Release Record Factually Correct.

Record Unchanged ≠ Record Originally Accurate.

Record Authenticated ≠ Every Claim Within the Record Validated.

Release Evidence Available ≠ Release Evidence Sufficient for Every

Lifecycle Decision.

Release Information Missing ≠ Release Condition Satisfied.

Record Uncertainty Acknowledged ≠ Record Uncertainty Resolved.

Release Record Corrected ≠ Original Release Event Changed.

Historical Record Inaccurate ≠ Historical Record May Be Silently Rewritten.

Release Record Superseded ≠ Previous Release Record Invalid for Historical

Reconstruction.

Release Record Version Increased ≠ Software Release Version Increased.

Current Release Record State ≠ Complete Historical Release State.

Record No Longer Operationally Needed ≠ Record Eligible for Deletion.

Release Record Stored ≠ Release Record Retrievable When Required.

Release Record Transferred ≠ Record Integrity and Accessibility

Automatically Preserved.

Release Record Classified as Complete ≠ Release Independently Validated as

Ready or Authorized.

Cross-Architecture Boundary

RRMM may consume governed references from:

GOA™ — Governance Architecture

AGA™ — Accountability Governance Architecture

SCS — Software Creation Standard

SDS — Software Dependency Standard

SCFS — Software Configuration Standard

SQS — Software Quality Standard

SSS — Software Security Standard

Within SRLS, RRMM receives release governance references from:

RCMM — Release Candidate Management Module

RRM — Release Readiness Module

RAM — Release Authorization Module

RRMM supports:

RWM — Release Withdrawal Module

SDPS — Software Deployment Standard

SCMS — Software Change Management Standard

SRTS — Software Retirement Standard

SLCS — Software Lifecycle Control Standard

RRMM preserves the release governance record and its historical continuity.

It does not replace the mechanisms responsible for establishing candidate identity, readiness, authorization, deployment, withdrawal, or lifecycle transitions.

Cross-Architecture Mechanism Principle

An operational mechanism may produce inputs for multiple governance architectures, but it belongs only to the architecture whose governed domain includes that mechanism itself. Other architectures govern only its domain-specific relevance or consequences.

RRMM may preserve evidence generated through software creation, configuration, dependency, quality, security, authorization, deployment, and operational processes.

It governs how that evidence is associated with and preserved within the software release record.

It does not redefine the underlying mechanisms or independently validate every referenced governance conclusion.

Canonical Boundary

RRMM governs how software release records are established, structured, associated, preserved, updated, verified, superseded, and maintained as traceable evidence of identifiable software releases and their applicable governance states throughout the software lifecycle.

It does not independently establish software readiness, grant release authorization, execute software deployment, withdraw releases, or certify software correctness, security, or operational acceptability.

Its principal governed output is the Software Release Record Profile.

Canonical Closing Statement

Software release governance cannot remain reliable if the evidence of what

was released, why it was considered ready, who authorized it, and under which conditions it was distributed cannot be reconstructed.

Modern software delivery increasingly involves distributed development

environments, automated build systems, external dependencies, AI coding agents, autonomous workflows, and multiple release channels.

These mechanisms can produce and distribute software rapidly.

But the speed of software delivery does not establish the completeness or integrity of its release history.

A software release may be technically successful while its record remains incomplete.

An authorization decision may be legitimate while its evidence is poorly preserved.

A release record may exist while the exact released artifact cannot be identified.

And a historical record may appear complete after modifications have silently erased the information originally recorded.

A release record is not reliable merely because it exists.

It must preserve identifiable relationships between the software product, release candidate, artifact composition, readiness assessment, authorization decision, release event, and applicable restrictions.

It must distinguish what was known at the time of release from what was discovered later.

It must preserve corrections without rewriting history.

And it must remain accessible and interpretable when software is deployed, modified, investigated, withdrawn, or retired.

Historical evidence must remain historical evidence—even when later knowledge changes the interpretation of the release.

This principle applies whether software is released by a human development team, an automated delivery pipeline, or an autonomous AI software factory.

The release mechanism may change.

The participating actors may change.

The software artifacts may evolve.

The requirement for reconstructable release governance remains.

RRMM transforms software release documentation into governed lifecycle

evidence — preserving not merely the claim that software was released, but the identifiable artifacts, decisions, conditions, events, and historical relationships required to establish what that release represented and how its governance state can be reconstructed over time.